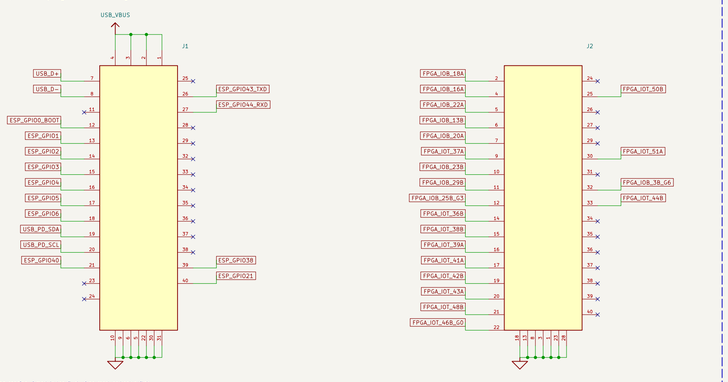
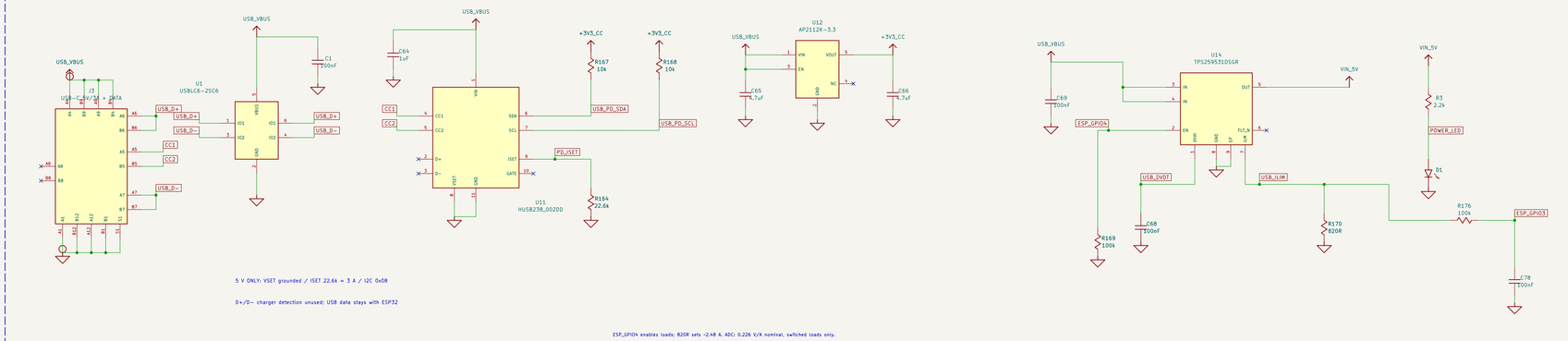
MODULE v1 / exact programmer mating interface



25 mm mating spacing / module top flush / no reset or boot switch

USB-C PD / 5 V, 3 A sink and protected power



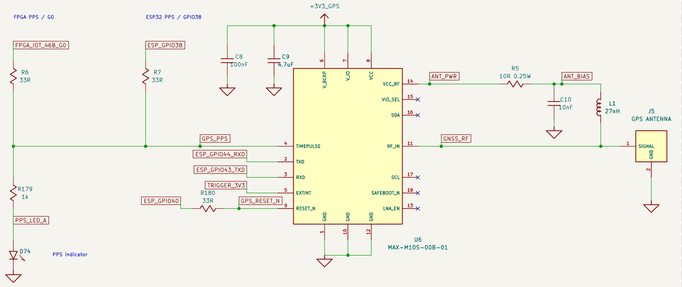
5 V ONLY: VSET grounded / ISET 22.6k = 3 A / I2C 0x08

D+/D- charger detection unused: USB data stays with ESP32

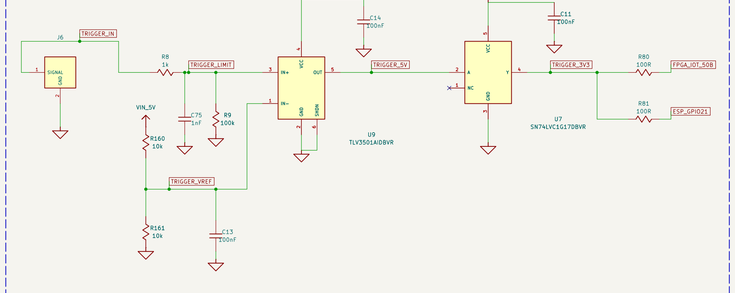
ESP_GPIO4 enables loads: R20R sets ~2.4A A; ADC: 0.226 V/A nominal, switched loads only.

Module and logic boot from raw 5 V. GPIO7/8 read PD status over I2C; firmware budgets current before enabling loads.

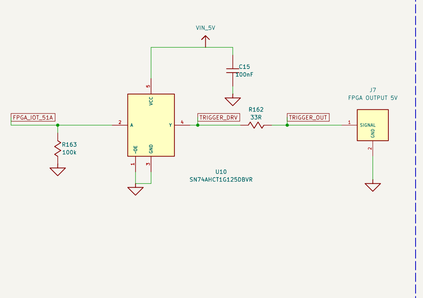
GNSS / PPS reference and active antenna bias



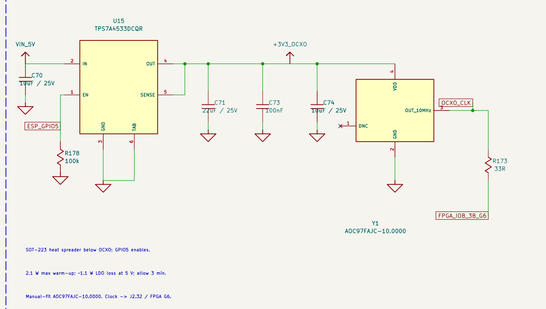
Trigger input / 5 V TTL / threshold ~2.5 V



Trigger output / 5 V TTL



10 MHz OCXO / 3.3 V linear supply

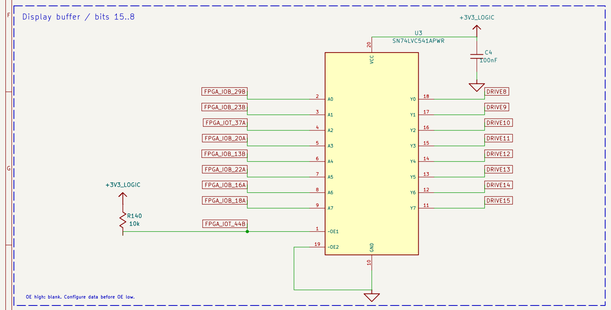


SOT-223 heat spreader below OCXO; GPIO5 enables.

2.1 W max warm-up; ~1.1 W LDO loss at 5 V: allow 3 min.

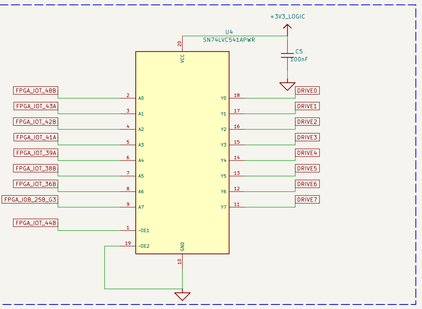
Manual-fit ASDC97FAJC-10.0000; Clock -> J2.32 / FPGA G6.

Display buffer / bits 15..8



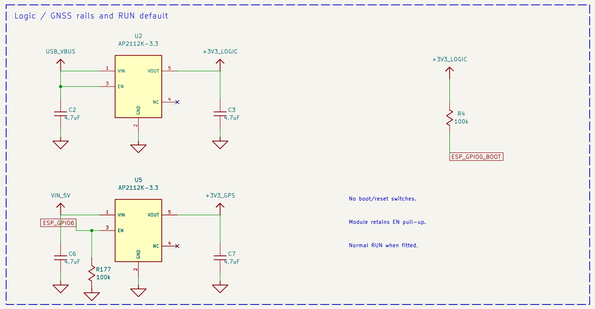
OE high: blank. Configure data before OE low.

Display buffer / bits 7..0



OE high: blank. Configure data before OE low.

Logic / GNSS rails and RUN default

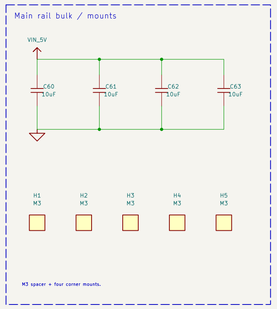


No boot/reset switches.

Module retains EN pull-up.

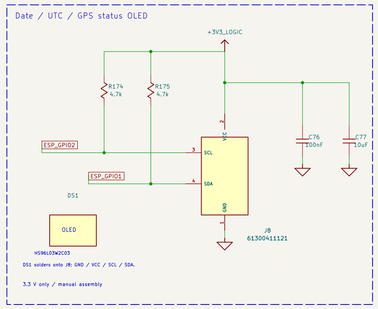
Normal RUN when fitted.

Main rail bulk / mounts



M3 spacer + four corner mounts.

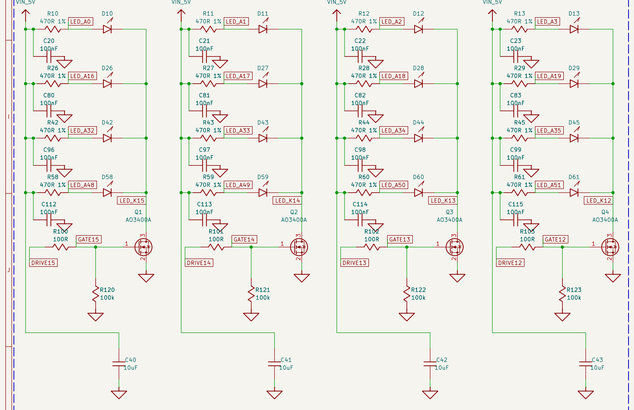
Date / UTC / GPS status OLED



DS1 solders onto J8: GND / VCC / SCL / SDA.

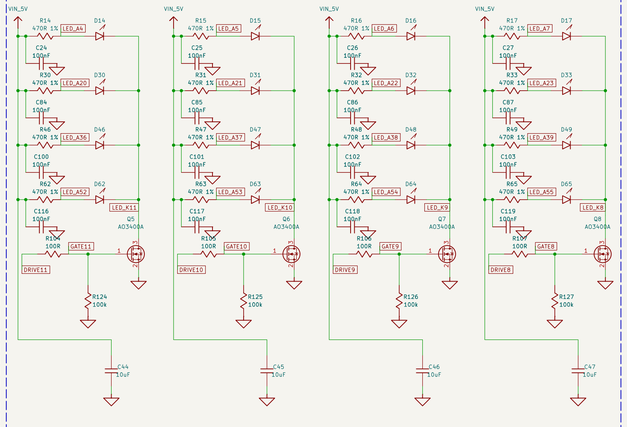
3.3 V only / manual assembly

OPTICAL BANK 1 / bits 15..12 / four identical rows



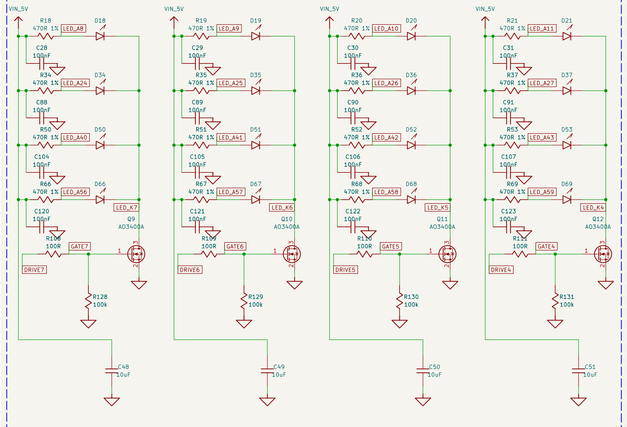
BIT 15 BIT 14 BIT 13 BIT 12

OPTICAL BANK 2 / bits 11..8 / four identical rows



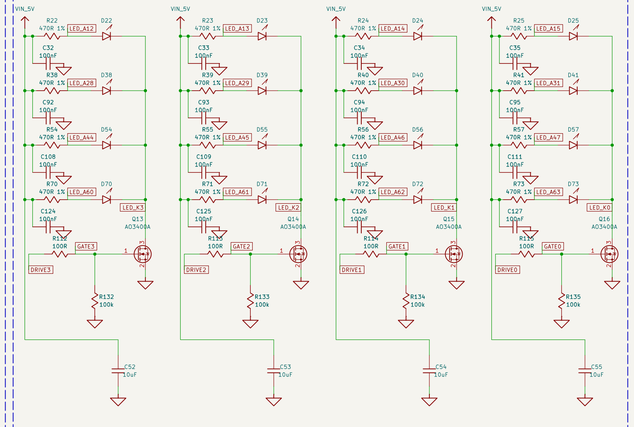
BIT 11 BIT 10 BIT 9 BIT 8

OPTICAL BANK 3 / bits 7..4 / four identical rows



BIT 7 BIT 6 BIT 5 BIT 4

OPTICAL BANK 4 / bits 3..0 / four identical rows



BIT 3 BIT 2 BIT 1 BIT 0

Diagnostic probe pads / functional names on PCB / bare copper, no assembly parts

TP1 // VBUS	TP2 // 5V_LOAD	TP3 // 3V3_LOGIC	TP4 // 3V3_CC	TP5 // 3V3_GPS	TP6 // 3V3_OCXO	TP7 // LOAD_EN	TP8 // OCXO_EN	TP9 // GPS_EN	TP10 // PD_SDA	TP11 // PD_SCL	TP12 // I_MON	TP13 // PPS_SRC	TP14 // PPS_FPGA		
USB_VBUS	VIN_5V	+3V3_LOGIC	+3V3_CC	+3V3_GPS	+3V3_OCXO	ESP_GPIO5	ESP_GPIO6	ESP_GPIO4	FPGA_IOT_50B	FPGA_IOT_51A	ESP_GPIO4	ESP_PPS	FPGA_IOT_44B_G8		
TP15 // PPS_ESP	TP16 // CLK_FPGA	TP17 // TRIG_FILT	TP18 // TRIG_DIR	TP19 // TRIG_CMP	TP20 // TRIG_5V5	TP21 // OUT_FPGA	TP22 // OUT_DRV	TP23 // GPS_TX	TP24 // GPS_RX	TP25 // ANT_PWR	TP26 // ANT_BIAS	TP27 // LED_DC_N			
ESP_GPIO38	FPGA_IOB_3B_G6	TRIGGER_LIMIT	TRIGGER_VREF	TRIGGER_5V	TRIGGER_3V3	FPGA_IOT_51A	TRIGGER_DRV	ESP_GPIO44_RXD	ESP_GPIO43_TXD	GPS_RESET_N	ANT_PWR	ANT_BIAS			
TP28 // OLED_SCL	TP29 // OLED_SDA	TP30 // BOOT_N													
ESP_GPIO22	ESP_GPIO21	ESP_GPIO0_BOOT													

bp-test
Sheet: /
File: calibrator.kicad_sch

Title: GPS optical time calibrator / module v1 carrier

Size: A1	Date:	Rev: A draft
KiCad E.D.A. 10.0.6		Id: 1/1